

Mining association patterns (Part 2)

milk, cheese and bread
are often bought together

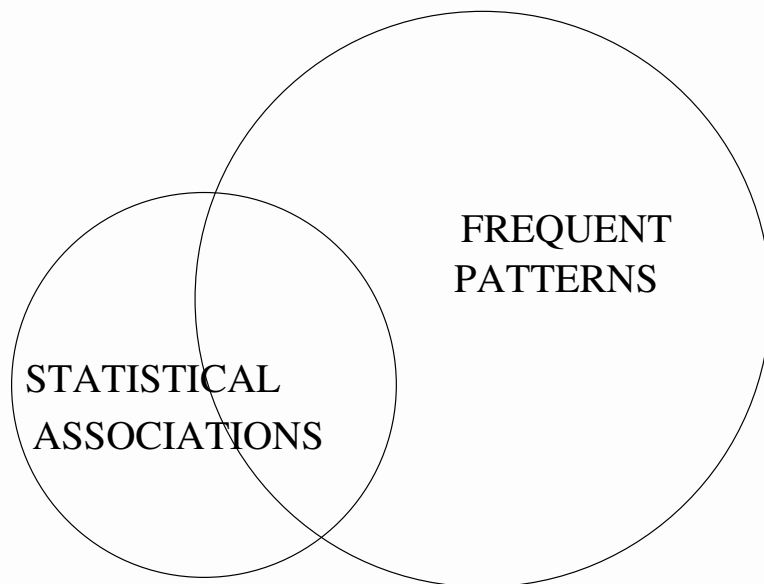
genes g1, g2, g3 and g4
are often over-expressed
in DLBC lymphomas

occurrence of certain insect species
makes it more likely to meet the
threatened white-backed woodpecker



How to find statistical association rules efficiently?

Problem: 2-step approach (frequents sets + postprocessing) very slow and often incomplete.



Recall experiments:

data	discovered %
Mushroom	100%
Chess	1%
T10I4D100K	100%
T40I10D100K	0%
Accidents	14%
Pumsb	27%
Retail	100%

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1. Properties of statistical association rules
2. Pruning the search space
3. Algorithms

1. Properties of statistical association rules

Goal: Find association rules $X \rightarrow C=c$ such that

1. rule expresses **statistical dependence**

$$P(X, C=c) > P(X)P(C=c)$$

2. discovery is **statistically significant** (not spurious/due to chance) $\Rightarrow$ likely to hold in future data
3. pattern is specialized only if improvement is statistically significant $\Rightarrow$ **overfitting avoidance**
4. (optional) pattern is **not otherwise specious** (misleading)
 - prune out associations that definitely do not present any causal relationship

Recall: Negative dependence between X and C = positive dependence between X and $\neg C$

	C	$\neg C$	Σ
X	$fr(XC) =$ $fr(X)P(C) + n\delta$	$fr(X\neg C) =$ $fr(X)P(\neg C) - n\delta$	$fr(X)$
$\neg X$	$fr(\neg XC) =$ $fr(\neg X)P(C) - n\delta$	$fr(\neg X\neg C) =$ $fr(\neg X)P(\neg C) + n\delta$	$fr(\neg X)$
Σ	$fr(C)$	$fr(\neg C)$	n

$\Rightarrow$ search rules of form $X \rightarrow C$ or $X \rightarrow \neg C$ expressing positive dependence between condition and consequent

$$\delta = \delta(X, C) = P(X, C) - P(X)P(C)$$

1.1 Could we use δ or γ for search?

1. Should we use δ or γ ?
 $\Rightarrow$ variable-based or value-based interpretation?
2. Is high δ or γ enough to guarantee good patterns?
3. How do we perform the search?
 - statistical dependence is **not a monotonic property!**
 - $AB \rightarrow C$ can express significant dependence, even if $A \rightarrow B$, $A \rightarrow C$ and $B \rightarrow C$ expressed independence ^a

^aVoluntary home task: invent an example where this holds!

Value-based and variable-based interpretation of association $X \rightarrow C = c$

Let I_X be an indicator variable:

$I_X = 1$, if X holds, and $I_X = 0$, otherwise.

Important: Are we interested in association between values $I_X=1$ and $C=c$ or between binary variables I_X and C ?

$\Rightarrow$ different goodness measures and different results!

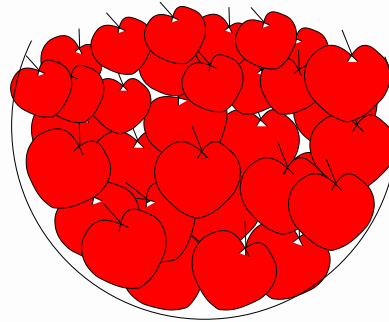
- **lift** γ measures strength of association between values
- **leverage** δ measures also strength of association between variables

Example: classifying apples by taste

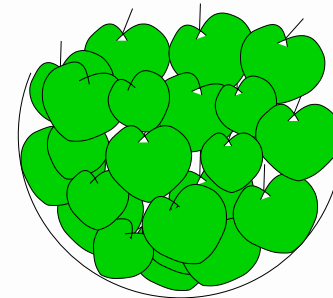
Which rule should you choose? It depends whether you want a strong association between values or variables!

red $\rightarrow$ sweet

$\delta=0.22$, $\gamma=1.67$



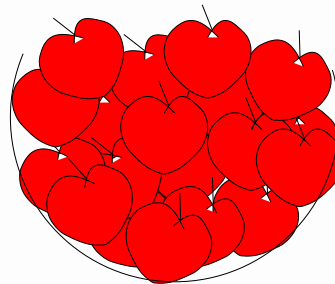
Basket 1
60 red apples
92% sweet



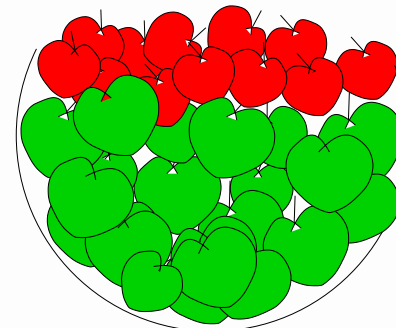
Basket 2
40 green apples
100% bitter

red $\wedge$ big $\rightarrow$ sweet

$\delta=0.18$, $\gamma=1.82$



Basket 1
40 large red apples
100% sweet



Basket 2
40 green + 20 small red apples
75% bitter

$n = 100$

$fr(\text{sweet})=55$

Is statistical dependence enough? Example

1000 students participated Mega Party. Data tells what each of them consumed in the party and what happened to them.

num	rule	fr_X	fr_C	fr_{XC}
1	<i>peppermint tea, sushi, chili sauce, sour cream → corona</i>	1	1	1
2	<i>vodka, sauerkraut, salmon → headache</i>	1	100	1
3	<i>cake → exam failure</i>	500	500	270
4	<i>magic mushrooms → intoxication</i>	20	20	20
5	<i>vodka → headache</i>	100	100	80
6	<i>vodka, salmon → headache</i>	40	100	30
7	<i>alcohol → exam failure</i>	333	500	300
8	<i>alcohol → cake</i>	333	500	200

Problem: High lift can be misleading

- a) 1 student got **corona** ($=C$)! The student was the only one who had combination $\mathbf{X}$ = **peppermint tea, sushi, chili sauce, sour cream**. $\gamma(\mathbf{X}, C) = \frac{P(\mathbf{X}C)}{P(\mathbf{X})P(C)} = \frac{1}{P(C)} = n$.
- b) 100 students had **headache** ($=C$), including one with unique combination $\mathbf{X}$ = **vodka, sauerkraut, salmon**. $\gamma(\mathbf{X}, C) = \frac{1}{P(C)} = \frac{1}{0.01} = 100$.
- Lift favors rare rules: $\gamma(\mathbf{X}, C) \leq \frac{1}{\max\{P(\mathbf{X}), P(C)\}}$

Problem: Leverage does not tell significance

Which rule is more significant?

- a) 500 students had **chocolate cake** (**X**) and 500 **failed the next day exam** (**C**), including 270 cake eaters.

$$\delta(\mathbf{X}, C) = \frac{1}{n^2}(n \cdot 270 - 500 \cdot 500) = 0.02.$$

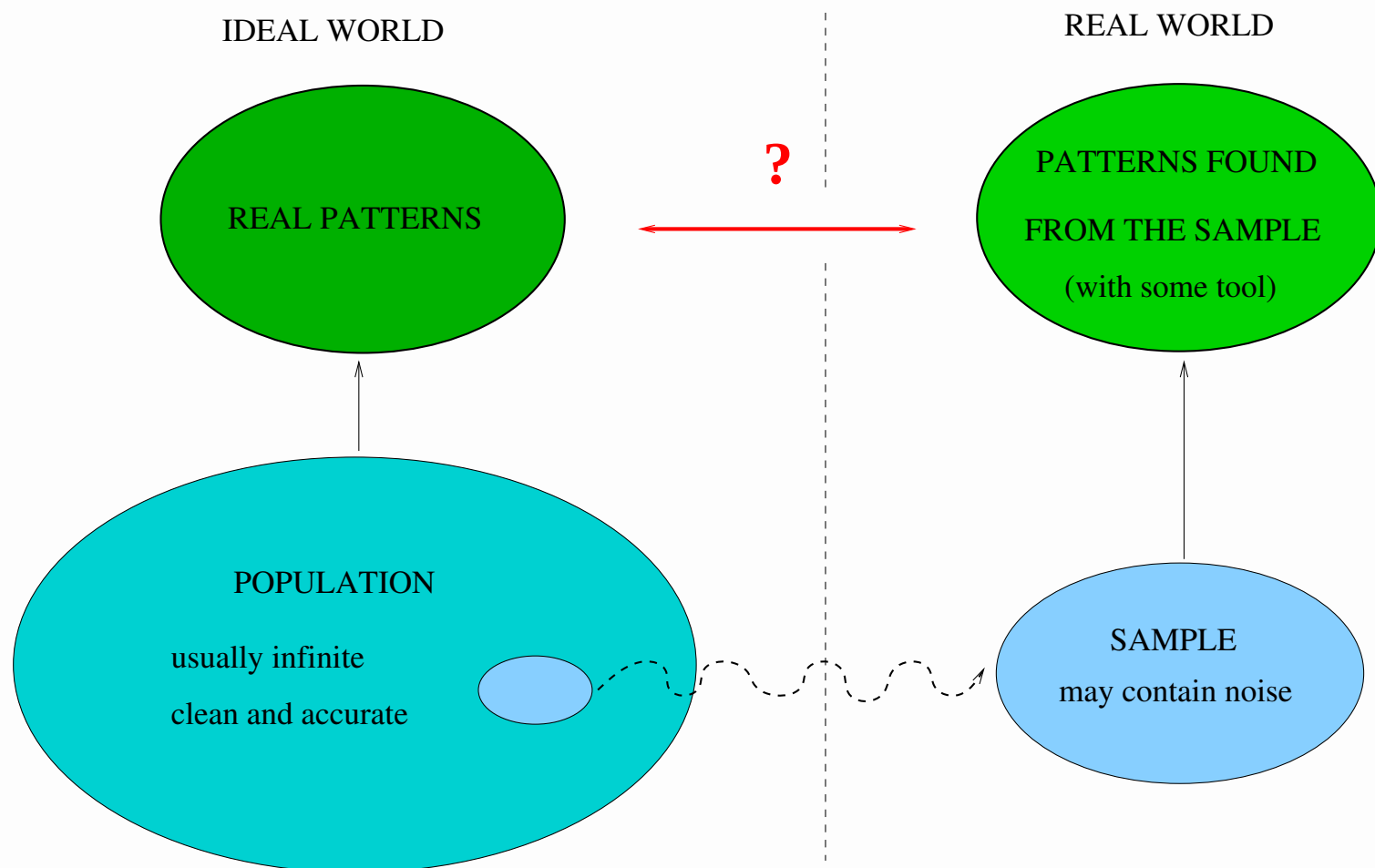
- b) 20 students tried **magic mushrooms** (**X**) and only them got a **serious intoxication** (**C**).

$$\delta(\mathbf{X}, C) = \frac{1}{n^2}(n \cdot 20 - 20 \cdot 20) = 0.0196.$$

- $\delta(\mathbf{X}, C) \leq \min\{P(\mathbf{X})P(\neg C), P(\neg \mathbf{X})P(C)\}$
- Leverage favors rules where $P(\mathbf{X}) \approx P(C) \approx 0.5$

⇒ What is the probability of observing such strong associations **by chance**?

1.2 Statistical significance: problem



Statistical significance

Idea: Given pattern, estimate probability of observing at least such a strong association, if the pattern actually expressed independence.

- if probability (p -value) is very small, the pattern is likely true
- many ways to estimate the probability
 - analytically (frequentist and Bayesian approaches)
 - empirically (randomization testing)

Further reading: Hämmäläinen & Webb (2019): A tutorial on statistically sound pattern discovery, Section 2.3

Significance measures for rules $X \rightarrow C=c$

1. Fisher's exact p -value:

$$p_F = \sum_{i=1}^J \frac{\binom{fr(\mathbf{X})}{fr(\mathbf{X}C=c)+i} \binom{fr(\neg\mathbf{X})}{fr(\neg\mathbf{X}C \neq c)+i}}{\binom{n}{fr(C=c)}},$$

where $J = \min\{fr(\mathbf{X}C \neq c), fr(\neg\mathbf{X}C=c)\}$

- very robust! use when possible
- often $\ln(p_F)$ more convenient in programs
- remember: small values good (for p_F and $\ln(p)$)

cake $\rightarrow$ exam failure $p_F = 6.8e-3$ ($\ln(p) = -5.0$)

mushrooms $\rightarrow$ intoxication $p_F = 3.0e-42$ ($\ln(p) = -95.6$)

Significance measures for rules $X \rightarrow C=c$

2. Mutual information:

$$MI = \log \frac{P(\mathbf{X}C)^{P(\mathbf{X}C)} P(\mathbf{X}\neg C)^{P(\mathbf{X}\neg C)} P(\neg\mathbf{X}C)^{P(\neg\mathbf{X}C)} P(\neg\mathbf{X}\neg C)^{P(\neg\mathbf{X}\neg C)}}{P(\mathbf{X})^{P(\mathbf{X})} P(\neg\mathbf{X})^{P(\neg\mathbf{X})} P(C)^{P(C)} P(\neg C)^{P(\neg C)}}$$

- now large values good
- you can get p -values for G statistic: $G = 2n \cdot MI$ (base e) or $G = 2n \cdot MI / \log_2(e)$ (base 2)

3. χ^2 -measure:

$$\chi^2 = \frac{n(P(\mathbf{X},C) - P(\mathbf{X})P(C))^2}{P(\mathbf{X})P(\neg\mathbf{X})P(C)P(\neg C)} = \frac{n\delta^2(\mathbf{X},C)}{P(\mathbf{X})P(\neg\mathbf{X})P(C)P(\neg C)}$$

- you can look p -values from the χ^2 distribution
- sensitive to accuracy of assumptions

Multiple hypothesis testing problem

Problem: The more patterns we test, the more will pass significance tests by chance.

E.g., if threshold $\alpha = 0.05$ and we test 10 000 spurious patterns, then about 500 patterns will incorrectly pass the test.

⇒ p -values need to be very small in DM!

⇒ Many strategies

Further reading: Hämmäläinen & Webb (2019): A tutorial on statistically sound pattern discovery, Section 6

1.3 Problem: Overfitted rules can be misleading

- Since $r5$ $\text{vodka} \rightarrow \text{headache}$ strong, no surprise that $r6$ $\text{vodka, salmon} \rightarrow \text{headache}$ is strong!
- If you knew only $r6$, you might think that vodka alone is safe!
- Possible that salmon and headache are **conditionally independent** given vodka or salmon may even prevent vodka headache (=negative conditional dependence)

Q and C are **conditionally independent**, given $X \Leftrightarrow$

$$P(Q, C|X) = P(Q|X)P(C|X) \Leftrightarrow$$

$$P(X, Q, C) = P(X, Q)P(C|X)$$

Problem: Specious associations are misleading

What about r3 **cake $\rightarrow$ failure** ($C \rightarrow F$)? ($fr = 270$)

- r7 **alcohol $\rightarrow$ failure** ($A \rightarrow F$) very strong, $P(F|A) = 0.9$ vs. $P(F|\neg A) = 0.3$
- If C independent of F given A , $P(F|CA) = 0.9$, so $CA \rightarrow F$ also strong (overfitted or redundant)
- If C independent of F given $\neg A$, $P(F|C\neg A) = 0.3$
- $fr(CA) = 200$ and $fr(C\neg A) = 300$
- Now expectation for $fr(CF) = fr(CAF) + fr(C\neg AF)$ is $fr(CA)P(F|A) + fr(C\neg A)P(F|\neg A) = 200 \cdot 0.9 + 300 \cdot 0.3 = 180 + 90 = 270$
- Rule **cake $\rightarrow$ failure** was just a side-product (specious)!

How to identify overfitted rules?

- $\mathbf{XQ} \rightarrow C$ can improve $\mathbf{X} \rightarrow C$ only, if $P(C|\mathbf{XQ}) > P(C|\mathbf{X})$
- If $P(C|\mathbf{XQ}) = P(C|\mathbf{X})$, then conditional independence
- What if $P(C|\mathbf{XQ})$ just slightly higher than $P(C|\mathbf{X})$?
(may be due to chance)
- $\Rightarrow$ test **statistical significance of improvement!**
 - Assume conditional independence and estimate the probability of the observed improvement (or more)
 - $\Rightarrow$ measures M_C for evaluating conditional dependence

When improvement is significant?

- **Value-based** associations: evaluate $M_C(\mathbf{XQ} \rightarrow C \mid \mathbf{X} \rightarrow C)$
- **Variable-based** associations more tricky!
- **problem:** adding $\mathbf{Q}$ to $\mathbf{X} \rightarrow C$ may improve $P(C|X)$, but worsen $P(\neg C|\neg \mathbf{X})$
 - $P(\text{Sweet}|\text{Red}) = 0.92 < 1.0 = P(\text{Sweet}|\text{Red}, \text{Big})$ but $P(\neg \text{Sweet}|\neg \text{Red}) = 1.0 > 0.75 = P(\neg \text{Sweet}|\neg(\text{Red}, \text{Big}))$
- evaluate $M_C(\mathbf{XQ} \rightarrow C \mid \mathbf{X} \rightarrow C)$ and $M_C(\neg \mathbf{X} \rightarrow \neg C \mid \neg(\mathbf{XQ}) \rightarrow \neg C)$

M = significance measure for unconditional dependence
 M_C = corresponding measure for conditional dependence

Lessons to learn

1. Importance of **statistical significance**
 - significant discoveries are likely to hold in future data
 - remember multiple hypothesis testing problem
2. Decide whether you need **variable-based or value-based** definition
 - choose right measures and tools
3. Overfitted (too specialized) rules can be misleading (**vodka, salmon → headache**)
 - but sometimes other types of rules may also be specious (**cake → exam failure**)

2. Pruning the search space

Problem: Statistical association and significance are **not monotone** properties!

- given goodness measure M reflecting strength or significance of association,
$$M(\mathbf{X} \rightarrow C=c) = f(n, fr(\mathbf{X}C=c), fr(\mathbf{X}), fr(C=c))$$
- M is **increasing by goodness** (ibg) if high values good (δ , γ , χ^2 , MI) or **decreasing by goodness** (dbg) if small values good (p_F , $\ln(p_F)$)
- if M ibg, we may have $M(\mathbf{XQ} \rightarrow C=c) > M(\mathbf{X} \rightarrow C=c)$
- **How to prune exponential search space?**

Basic idea: branch and bound search using monotonic upper or lower bounds for M

Idea: In set X evaluate monotonic upper or lower bounds for $M(XQ \Rightarrow C)$

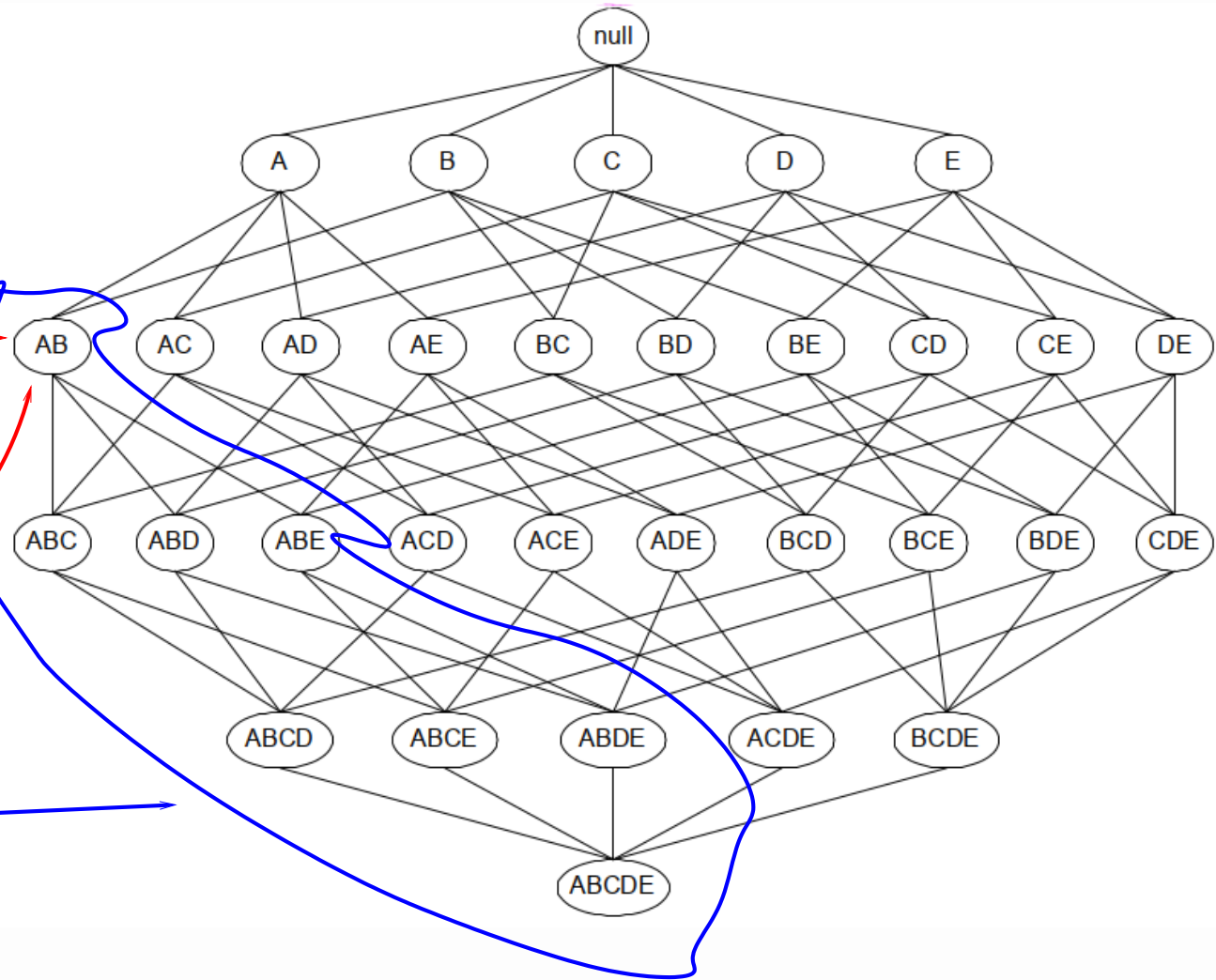
2 pruning rules:

1) find out here that consequent C cannot yield any good rules in AB or its extensions $\Rightarrow$

disable consequent C here

2) find out here that no consequent can yield good rules in AB or its extensions $\Rightarrow$

prune all sets here



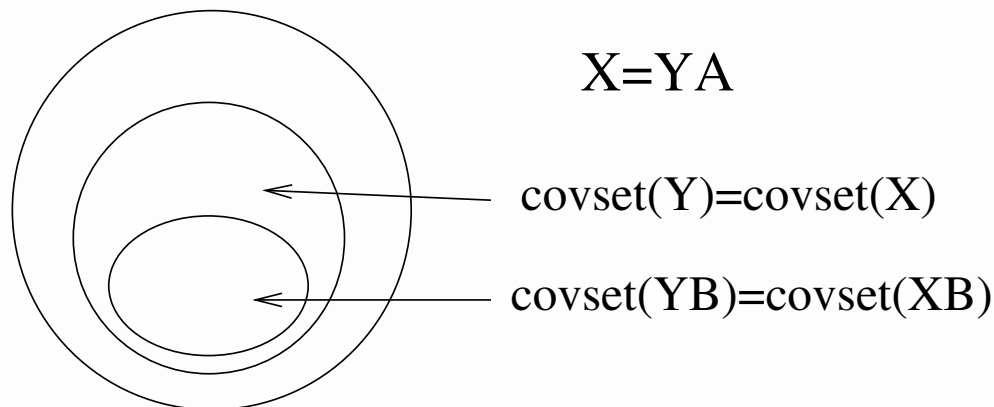
Pruning methods

- If M ibg, estimate upperbounds for $M(\mathbf{XQ} \rightarrow C=c)$ using information in $\mathbf{X}$ (possibly $\mathbf{X} = \emptyset$) ($P(C)$ known)
- e.g., upperbounds for δ :
 - i) $\delta(\mathbf{X} \rightarrow C) \leq P(C)P(\neg C)$ for any $\mathbf{X}$
 - ii) $\delta(\mathbf{XQ} \rightarrow C) \leq P(\mathbf{X})P(\neg C)$ for any $\mathbf{Q}$
 - iii) $\delta(\mathbf{XQ} \rightarrow C) \leq P(\mathbf{XC})P(\neg C)$ for any $\mathbf{Q}$
- If M is dbg, estimate lowerbounds
- Search for only top- Q rules $\Rightarrow$ update threshold $\max_M$ or $\min_M$ when new good rules found
- Estimate in advance which rules would be overfitted and can be ignored

Pruning methods

- Utilize minimality condition: If $\mathbf{X} = \mathbf{Y} \cup \{A\}$ and $P(A=a|\mathbf{Y}) = 1.0$ (i.e., $P(\mathbf{Y}A=a) = P(\mathbf{Y})$), then
 - i) A and $\neg A$ can be ignored in all $\mathbf{XQ}$
rule $\mathbf{YQ} \rightarrow A=a$ redundant vs. $\mathbf{Y} \rightarrow A=a$
 - ii) all $B=b$, $B \notin \mathbf{X}$, can be ignored in all $\mathbf{XQ}$
e.g., $\mathbf{X} \rightarrow B$ redundant vs. $\mathbf{Y} \rightarrow B$

$\text{covset}(A)$ = transactions covered by A



In the figure, $\text{covset}(\mathbf{Y}) \subseteq \text{covset}(A)$ since $P(A|\mathbf{Y}) = 1$

3. Algorithms for statistical association rules

- **Magnum Opus** ^a: search for sufficiently strong rules with γ or δ and test significance of improvement with a hypergeometric test
 - value-based: compares only $\mathbf{XQ} \rightarrow C$ vs. $\mathbf{X} \rightarrow C$
- **Kingfisher** ^b: search for the most significant rules with a significance measure (Fisher's p_F , χ^2 , MI , etc.)
 - variable-based: compares both $\mathbf{XQ} \rightarrow C$ vs. $\mathbf{X} \rightarrow C$ and $\neg(\mathbf{XQ}) \rightarrow \neg C$ vs. $\neg\mathbf{X} \rightarrow \neg C$

^aWebb 2005, Webb 2007

^bHämäläinen 2012

Algorithms for statistical classification rules

Now the consequence is fixed! Methods for finding best rules $\mathbf{X} \rightarrow C$ with

- χ^2 (Morishita and Sese, 2000^a and Nijssen and Kok, 2006)
- with any convex measure (like χ^2 , MI), when $\mathbf{X}$ is closed (Nijssen et al., 2009)
- with measures of strength like δ and γ (Li, 2006)
- etc.

^aalso show that the problem of finding the best classification rule with χ^2 is *NP*-hard

Example: Kingfisher algorithm

Problem: Given a set of binary attributes $\mathbf{R} = \{A_1, \dots, A_k\}$. Search for the most significant positive and negative dependency rules $\mathbf{X} \rightarrow A=a$, where $\mathbf{X} \subsetneq \mathbf{R}$, $A \in \mathbf{R} \setminus \mathbf{X}$, $a \in \{0, 1\}$!

- significance measures: p_F , $\ln(p_F)$, χ^2 , MI , z -score
- in the following, let's use p_F
- rules are **non-redundant** ($\mathbf{X}$ contains no extra attributes which do not improve the dependency):
 $\nexists \mathbf{Y} \subsetneq \mathbf{X}$ such that $p_F(\mathbf{Y} \rightarrow A=a) \leq p_F(\mathbf{X} \rightarrow A=a)$
 - remember: smaller p -values better

Algorithm: the main idea

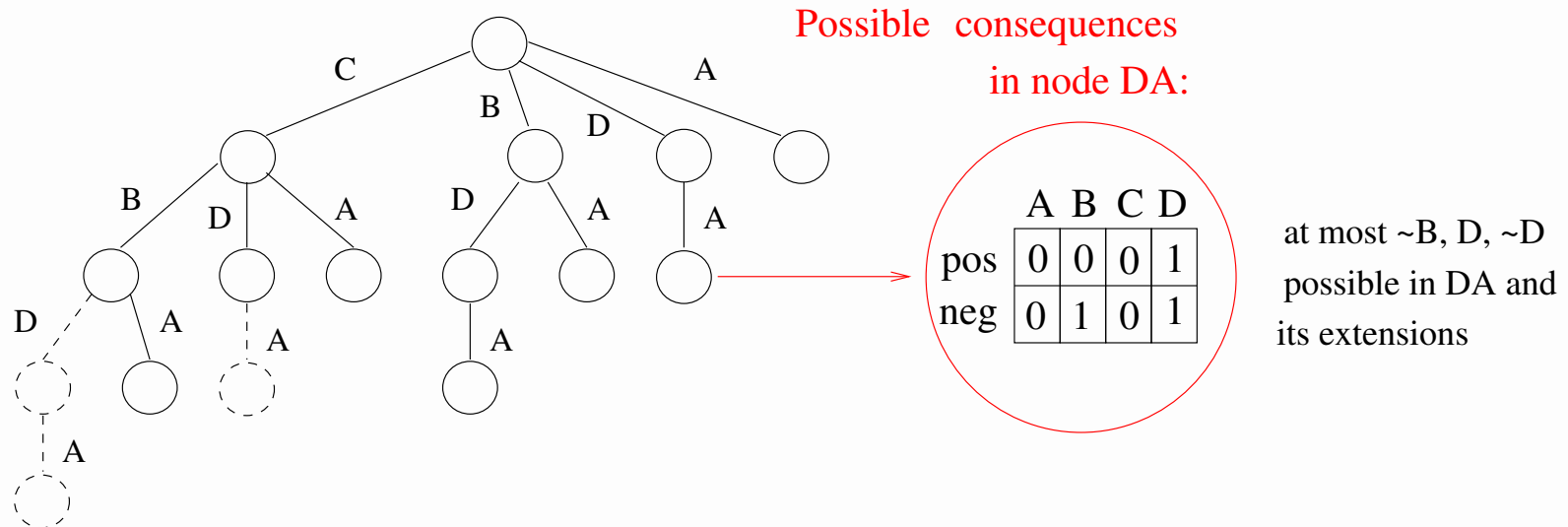
≈ Boosted branch-and-bound

- Generate an enumeration tree (breadth-first) and keep record on possible consequences at each node
- Consequence $A_i=a_i$ is set **impossible** in set $\mathbf{X}$, if for all sets $\mathbf{Q}$, rule $\mathbf{X} \setminus \{A_i\}\mathbf{Q} \rightarrow A_i=a_i$ is insignificant or redundant
- Consequence can be possible **only if it was possible in all parents** (monotonicity!)
- A node is pruned when no possible consequences left

Algorithm: the main idea

In each node

- bitvectors for possible positive and negative consequences
- numeric vector for best p_F -values of simpler rules
 - for checking redundancy



How to determine possible consequences?

At each node (set) $\mathbf{X}$

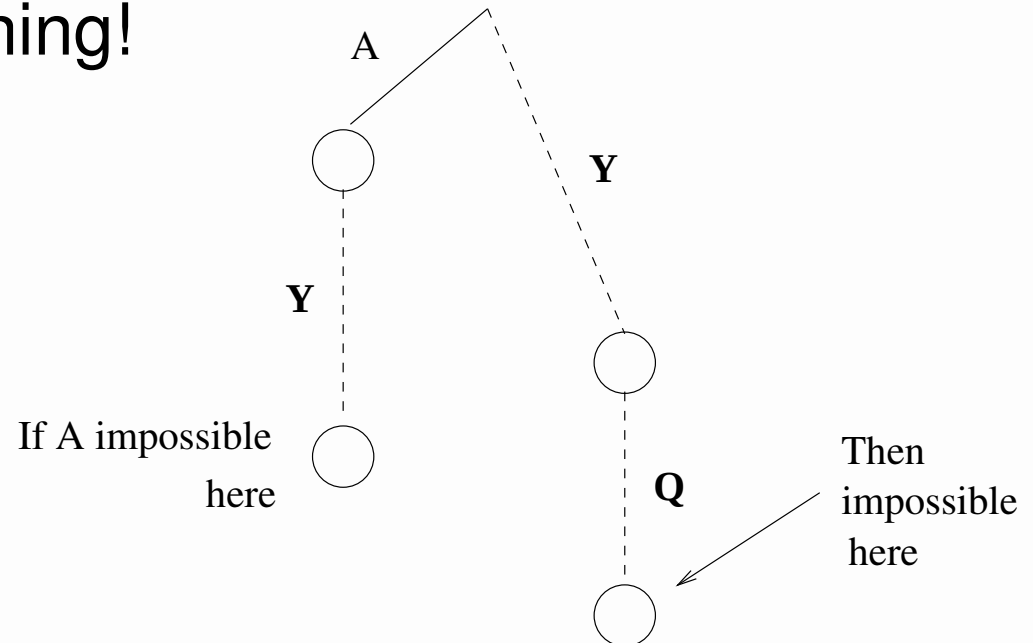
1. Initialization: combine parents' "possible" vectors (bit-and)
2. Updating: estimate lower bounds (LB) for $p_F(\mathbf{X} \setminus \{A_i\} \mathbf{Q} \rightarrow A_i=a_i)$ and decide if $A_i=a_i$ impossible in node $\mathbf{X}$
 - both for $A_i \in \mathbf{X}$ and $A_i \notin \mathbf{X}$
 - uses 3 different lower bounds $LB1$, $LB2$, $LB3$
 $LB1$: only $fr(A_i=a_i)$ known; $LB2$: $fr(A_i=a_i)$ and $fr(\mathbf{X})$ known, $A_i \notin \mathbf{X}$; $LB3$: $fr(A_i=a_i)$ and $fr(\mathbf{X})$ known, $A_i \in \mathbf{X}$
3. Utilize minimality condition

Algorithm: possible consequences

4. Lapis philosophorum principle (LP):

If $A=a$ impossible in X , then set it impossible in parent Y , $X = Y \cup \{A\}$, and all its descendants YQ

- Note: Y was already processed at previous level, but now updated again
- Very efficient pruning!



Extra: Simulation

Data

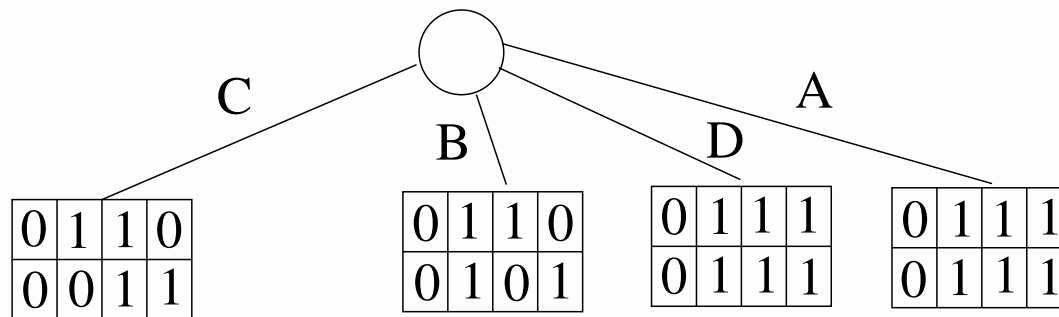
$$\mathbf{R} = \{A, B, C, D\}$$

$$n = 100$$

set	freq.
$ABC\neg D$	10
$A\neg B\neg CD$	85
$\neg AB\neg CD$	5

Search for the 10 best rules from the example data, when initial $p_{max} = 1.2 \cdot 10^{-8}$.

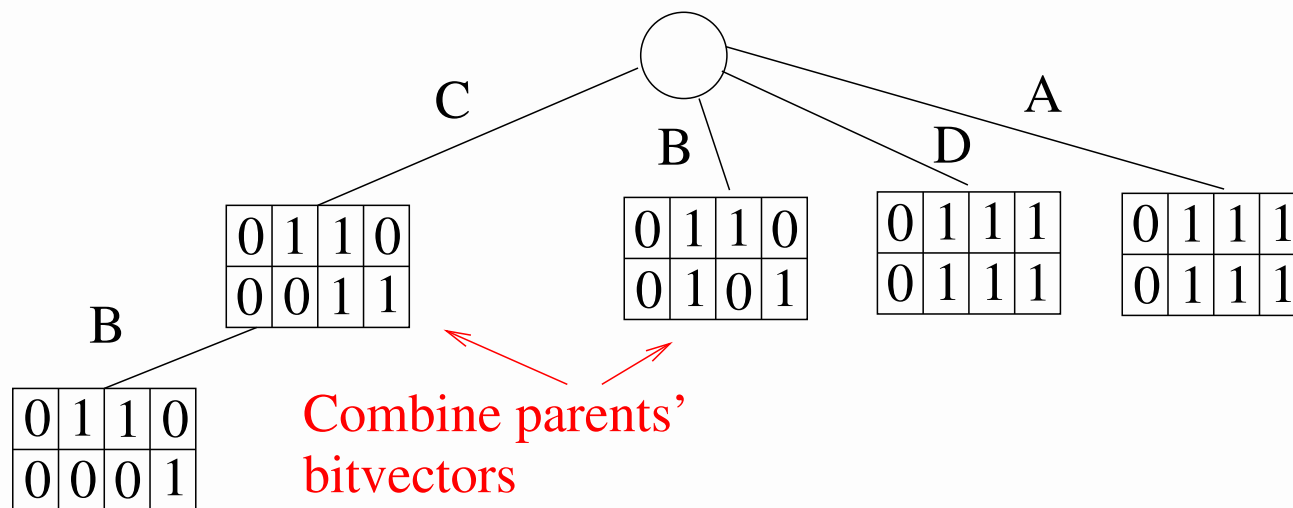
Simulation level 1: use LBs



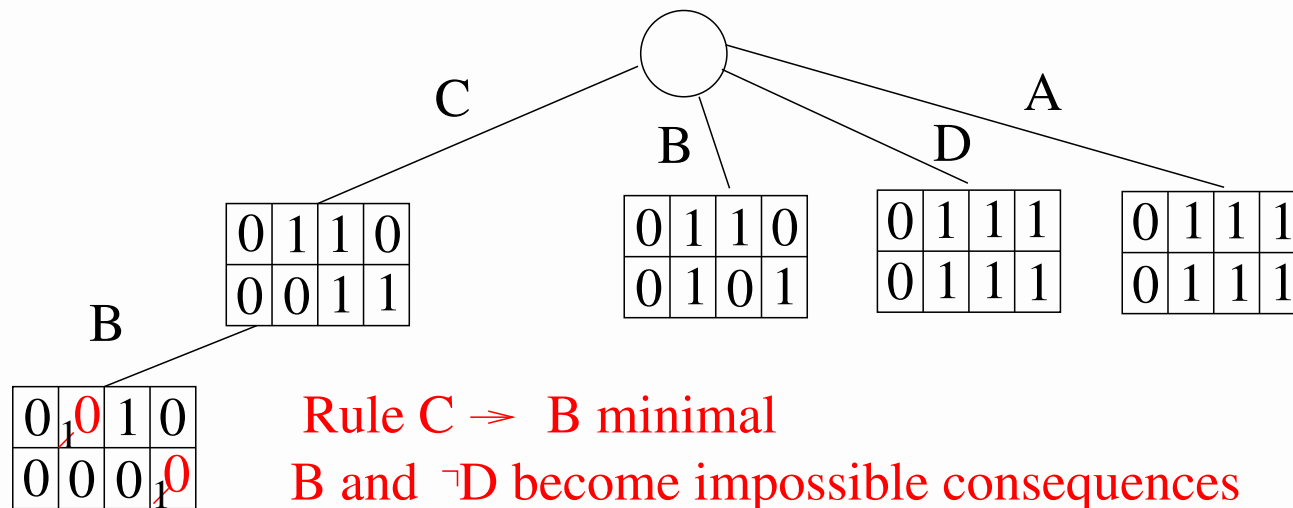
LB1: A and $\neg A$ are impossible consequences

Otherwise, possible consequences are determined by LB2

Simulation level 2: initialize CB



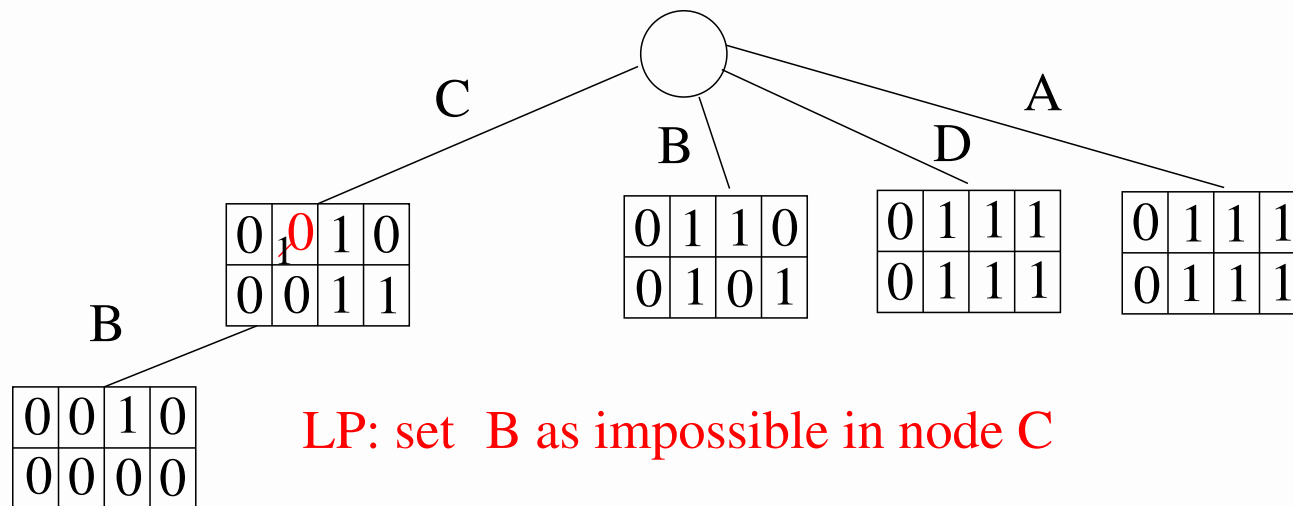
Simulation level 2: evaluate CB



Rule $C \Rightarrow B$ minimal
 B and $\neg D$ become impossible consequences
 (more specific rules would be redundant)

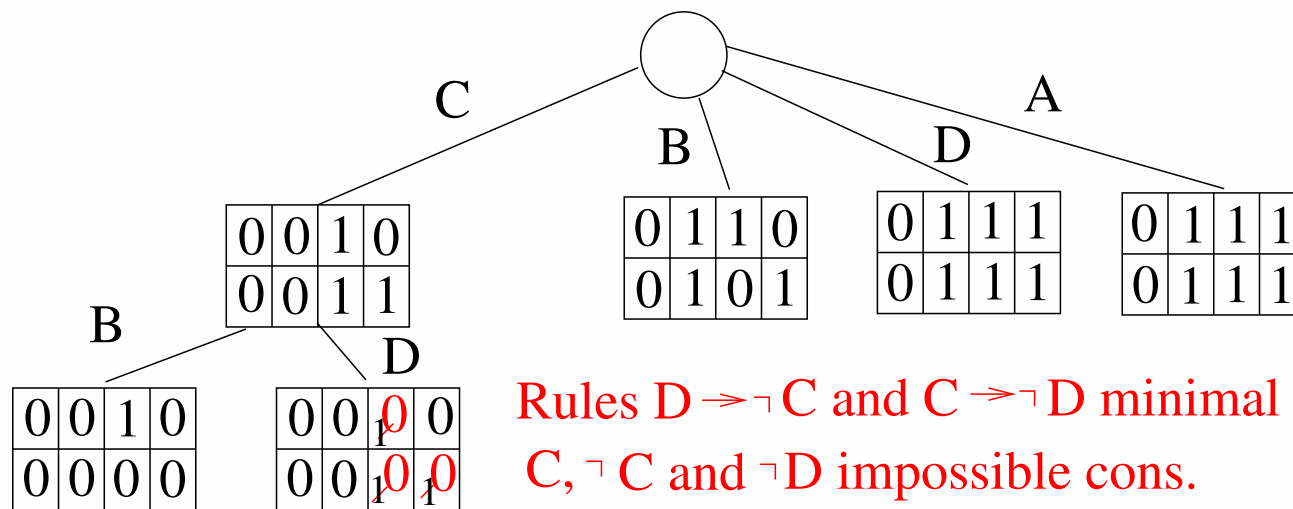
Rule	p
$C \rightarrow B$	$1.7 \cdot 10^{-10}$

Simulation level 2: utilize LP



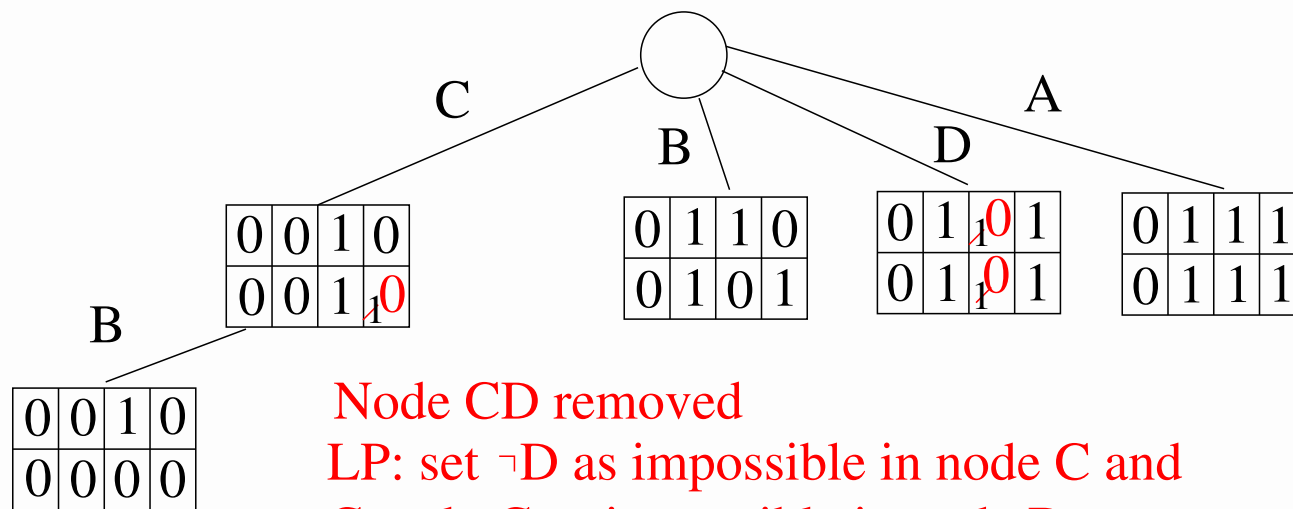
Rule	p
$C \rightarrow B$	$1.7 \cdot 10^{-10}$

Simulation level 2: create and evaluate CD



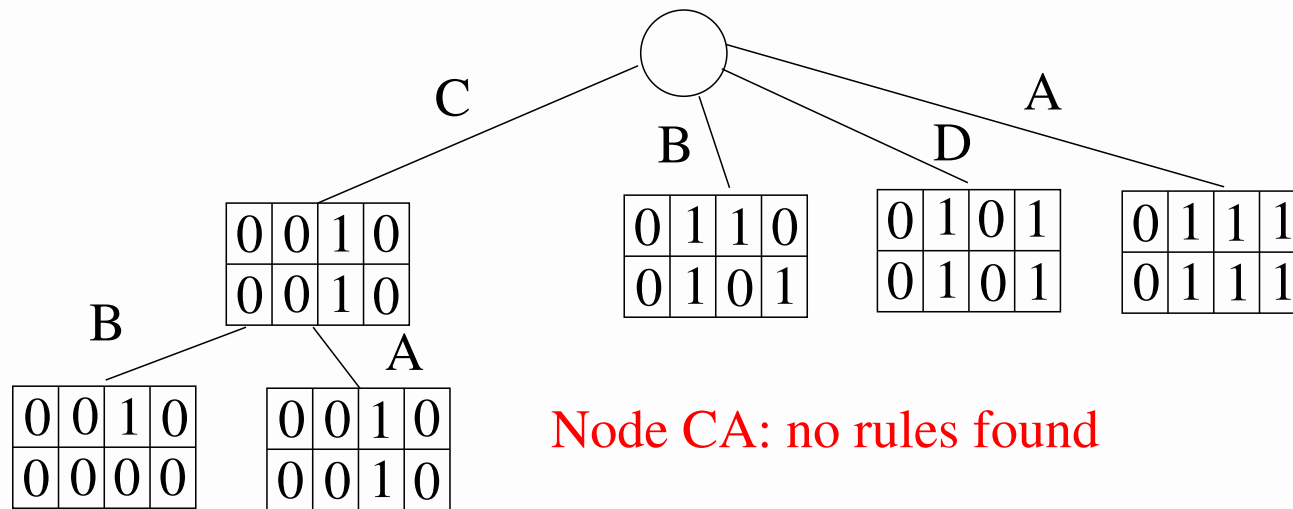
Rule	p
$D \rightarrow \neg C$	$5.8 \cdot 10^{-14}$
$C \rightarrow B$	$1.7 \cdot 10^{-10}$

Simulation level 2: utilize LP



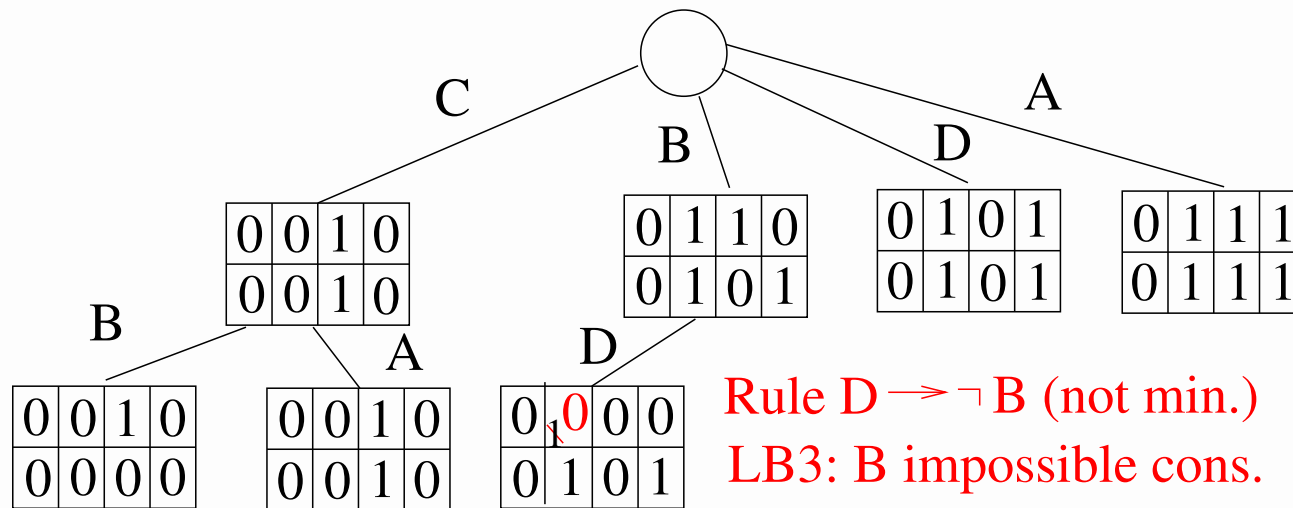
Rule	p
$D \rightarrow \neg C$	$5.8 \cdot 10^{-14}$
$C \rightarrow B$	$1.7 \cdot 10^{-10}$

Simulation level 2: evaluate CA



Rule	p
$D \rightarrow \neg C$	$5.8 \cdot 10^{-14}$
$C \rightarrow B$	$1.7 \cdot 10^{-10}$

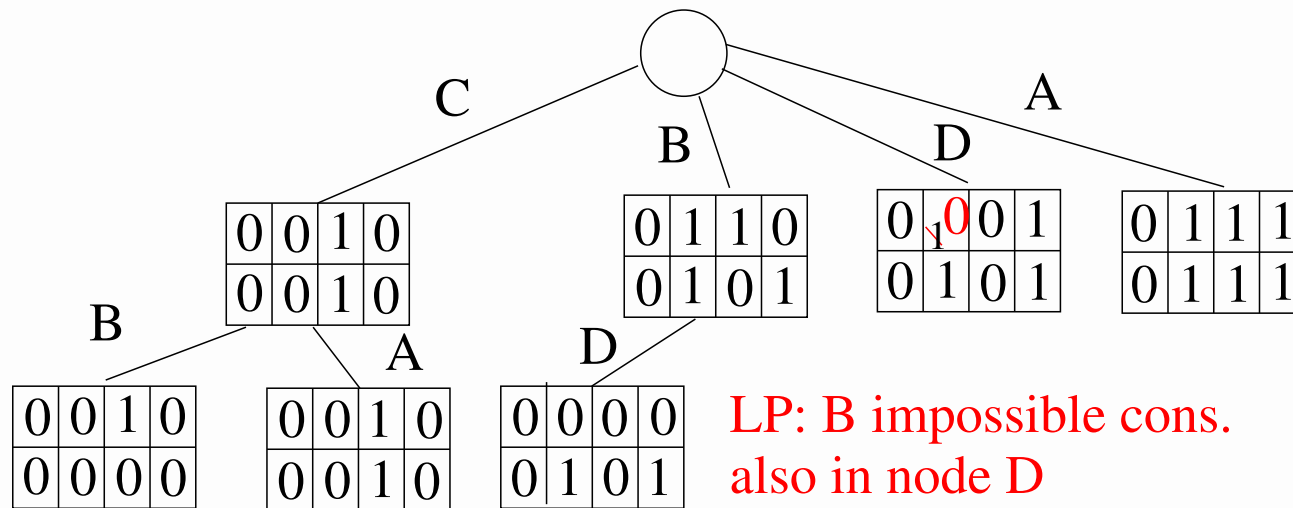
Simulation level 2: evaluate BD



Rule $D \rightarrow \neg B$ (not min.)
 LB3: B impossible cons.

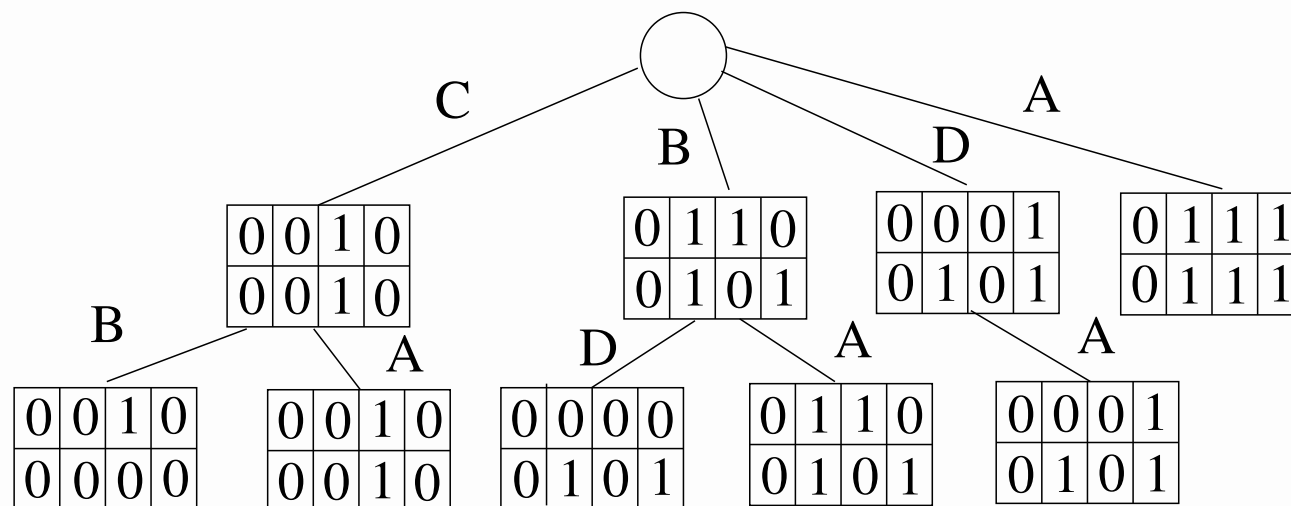
Rule	p
$D \rightarrow \neg C$	$5.8 \cdot 10^{-14}$
$C \rightarrow B$	$1.7 \cdot 10^{-10}$
$D \rightarrow \neg B$	$1.7 \cdot 10^{-10}$

Simulation level 2: utilize LP



Rule	p
$D \rightarrow \neg C$	$5.8 \cdot 10^{-14}$
$C \rightarrow B$	$1.7 \cdot 10^{-10}$
$D \rightarrow \neg B$	$1.7 \cdot 10^{-10}$

Simulation level 2: evaluate BA and DA

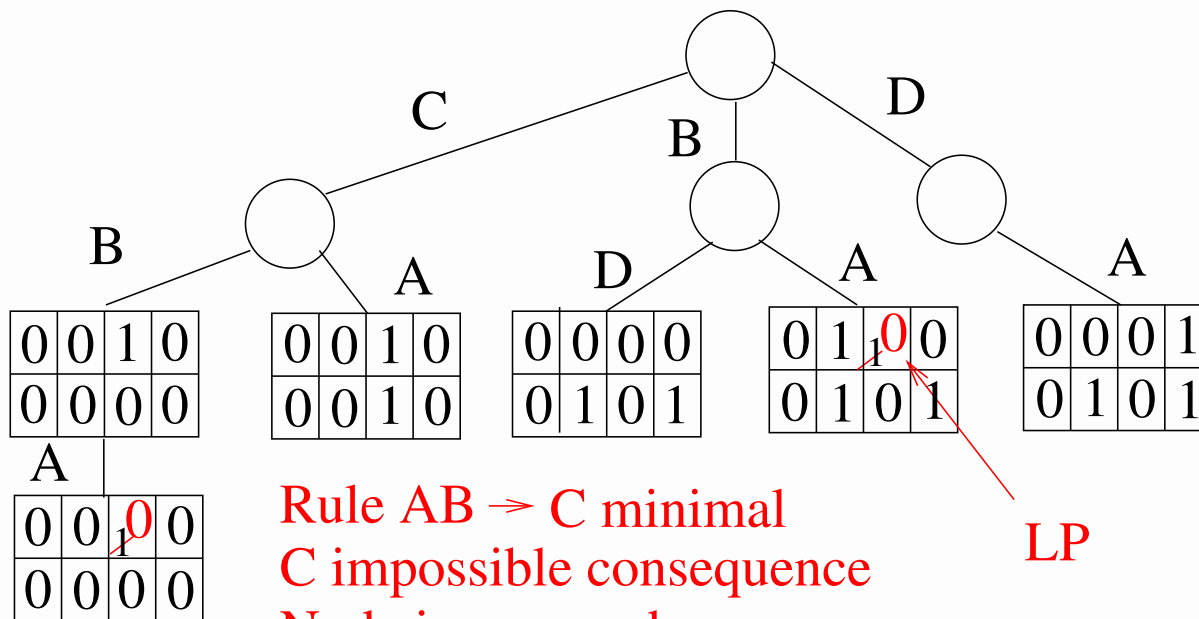


Rule	p
$D \rightarrow \neg C$	$5.8 \cdot 10^{-14}$
$C \rightarrow B$	$1.7 \cdot 10^{-10}$
$D \rightarrow \neg B$	$1.7 \cdot 10^{-10}$

BA and DA: no rules found

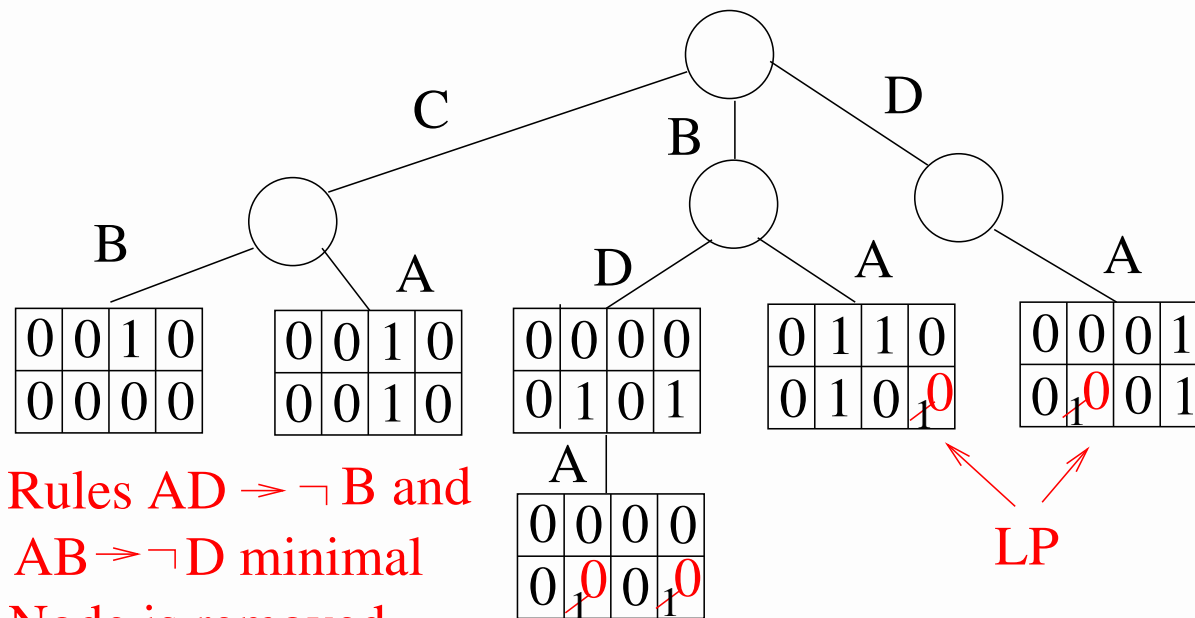
Note: Set A (1st level) no more needed.

Simulation level 3: evaluate CBA + use LP



Rule	p
$D \rightarrow \neg C$	$5.8 \cdot 10^{-14}$
$AB \rightarrow C$	$5.8 \cdot 10^{-14}$
$C \rightarrow B$	$1.7 \cdot 10^{-10}$
$D \rightarrow \neg B$	$1.7 \cdot 10^{-10}$

Simulation level 3: evaluate BDA + use LP



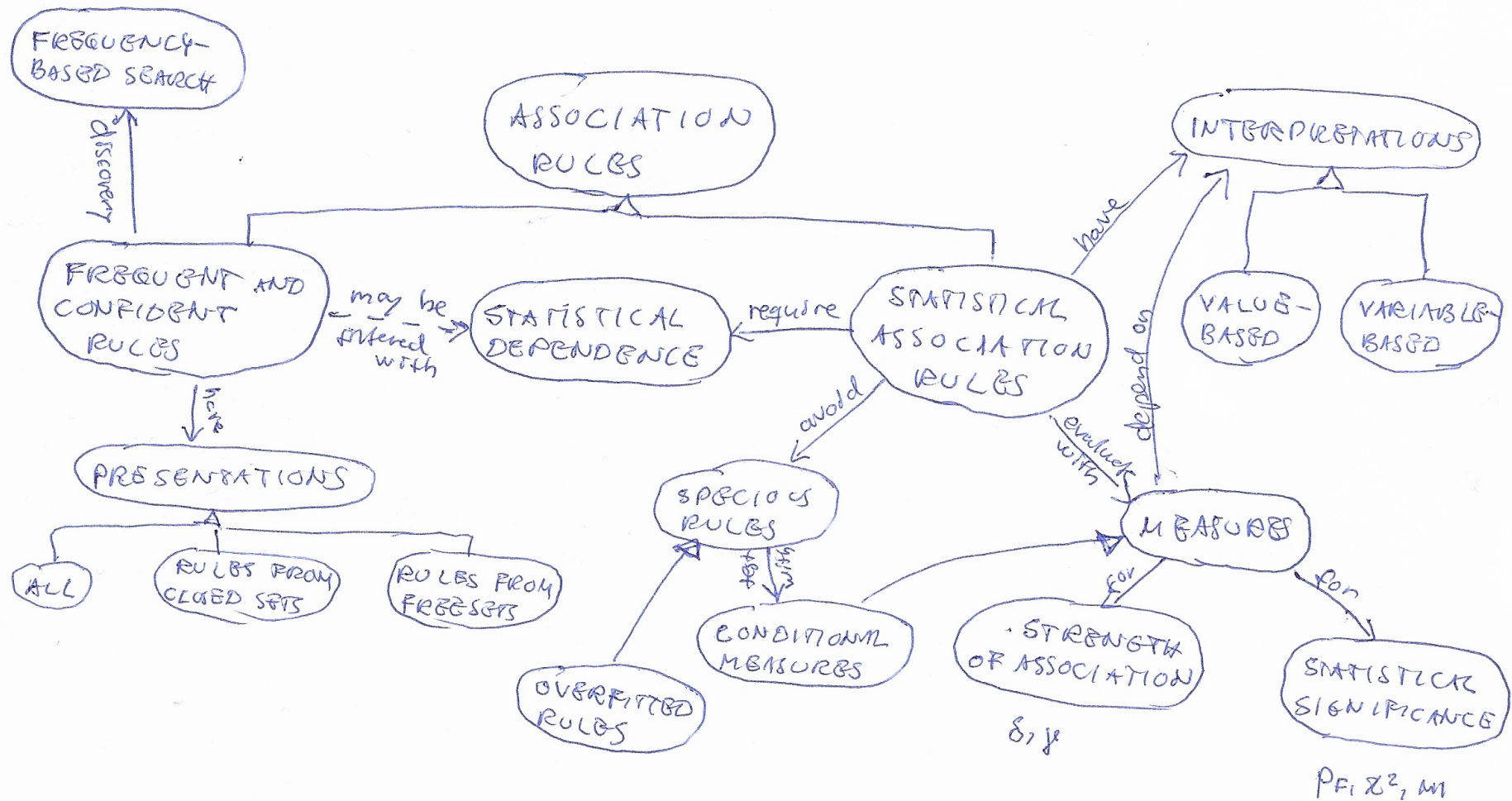
Rule	p
$AD \rightarrow \neg B$	$3.9 \cdot 10^{-18}$
$D \rightarrow \neg C$	$5.8 \cdot 10^{-14}$
$AB \rightarrow C$	$5.8 \cdot 10^{-14}$
$AB \rightarrow \neg D$	$5.8 \cdot 10^{-14}$
$C \rightarrow B$	$1.7 \cdot 10^{-10}$
$D \rightarrow \neg B$	$1.7 \cdot 10^{-10}$

DONE!

Summary

- If you want statistical associations, search them directly! (postprocessing frequent sets causes false positives & false negatives)
- statistical dependence **not monotonic** property (may be $AB \rightarrow C$ even if $A \perp\!\!\!\perp B$, $A \perp\!\!\!\perp C$, $B \perp\!\!\!\perp C$)
- **Secret:** $UB(M)$ or $LB(M)$ may behave monotonically!
 - $\forall Q$: if $UB(M(\mathbf{XQ} \rightarrow C)|fr(c)) < \min_M$,
 $UB(M(\mathbf{XQ} \rightarrow C)|fr(c), fr(\mathbf{X})) < \min_M$, or
 $UB(M(\mathbf{XQ} \rightarrow C)|fr(c), fr(\mathbf{X}), fr(\mathbf{XC})) < \min_M$, $\mathbf{XQ} \rightarrow C$ can be pruned! (here M ibg)
- Remember overfitted and other misleading rules!

Concept map of Association rules



Reading

- Hämmäläinen and Webb: A tutorial on statistically sound pattern discovery. Data Mining and Knowledge Discovery 33(2):325-377, 2019. **Sections 3.1, 4.1, 4.4.**
- Properties of statistical association rules explained with the Mega Party example (in MyCourses)
- **(Optional)** Hämmäläinen: Kingfisher: an efficient algorithm for searching for both positive and negative dependency rules with statistical significance measures, Knowledge and Information Systems 32: 383-414, 2012. **Sections 1-3, 4.1-4.2, the main idea from 5.1.**

Other references

- Li: On optimal rule discovery. IEEE Transactions on Knowledge and Data Engineering, 18(4):460-471, 2006.
- Lindley and Novick: The Role of Exchangeability in Inference, Annals of Statistics 9(1):45-58, 1981.
- Morishita and Sese: Traversing itemset lattices with statistical metric pruning. ACM SIGMOD-SIGACT-SIGART symposium on Principles of database systems, 2000.
- Nijssen and Kok: Multi-class correlated pattern mining. 4th International Workshop on Knowledge Discovery in Inductive Databases, 2006.

Other references

- Nijssen, Guns, and De Raed. Correlated itemset mining in ROC space: a constraint programming approach. 15th ACM SIGKDD conf. Knowledge Discovery and Data Mining, 2009.
- Webb: Discovering significant patterns. Machine Learning, 68(1):1-33, 2007.
- Webb: Magnum Opus. Software, G. I. Webb & Associates, Melbourne, Australia.
- Webb and Zhang. K-optimal rule discovery. Data Mining and Knowledge Discovery, 10(1), 2005.